Vitor Gama

+1-304-435-5900 | @ vrg00001@mix.wvu.edu | vitor-renan-gama | Morgantown, WV, USA

Education

West Virginia University — Morgantown, WV, USA
Ph.D. student, Chemical Engineering; — *Aug. 2021 – Dec. 2025 (Expected)*
GPA: 3.75/4.00

Federal University of Campina Grande — Campina Grande, Paraiba, Brazil
M.Sc., Chemical Engineering; — *Sept. 2018 – April 2021*
GPA: 9.14/10.00

Federal University of Campina Grande — Campina Grande, Paraiba, Brazil
B.Sc., Chemical Engineering; — *May 2013 – Aug. 2018*
GPA: 8.18/10.00

Awards & Achievements

– **AIChE Environmental Division Graduate Paper Award (2024):** Awarded 2nd place with the paper titled "Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO_2 Production".

– **Jack and Marietta Mullenger Fellowship (2024)**

– **Statler College PhD Recruitment Fellowship (2021)**

Research Publications

Complete list available on my Google Scholar.

[1] **Vitor Gama**, Deepanjali Roy, Fernando V. Lima and Oishi Sanyal *"Connecting Material Characteristics with System Properties for Membrane-based Direct Air Capture (m-DAC) using Process Operability and Inverse Design Approaches"*, **Industrial Engineering and Chemistry Research**. *(under review)*.

[2] **Vitor Gama**, Beatriz Dantas, Oishi Sanyal, and Fernando V. Lima. *"Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO_2 Production"*, **ACS Engineering Au**, 4.4 (Aug. 2024), pp. 394-404. DOI: 10.1021/acsengineeringau.3c00069. Publisher: American Chemical Society.

[3] **Vitor V. Gama**, San Dinh, Victor Alves, Beatriz N. A. Dantas, Brent A. Bishop, and Fernando V. Lima. *"Modeling and Process Operability Analysis of a Direct Air Capture System"*, **IFAC-PapersOnLine**, 13th IFAC Symposium on Dynamics and Control of Process Systems, including Biosystems DYCOPS 2022, Vol. 55.7 (Jan. 2022), pp. 316-321. DOI: 10.1016/j.ifacol.2022.07.463. ISSN: 2405-8963.

Conference Presentations

[1] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Leveraging Process Operability Mapping to Support Experimental Membrane Direct Air Capture (m-DAC) Solutions.* 2024 AIChE Annual Meeting, San Diego, CA, October 29, 2024 (Oral Presentation)

[2] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO_2 Production.* Membranes: Materials and Processes – Gordon Research Conference, New London, NH, July 28 – August 2, 2024 (Poster Presentation)

[3] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Evaluation of Hollow Fiber Membrane Modules for CO_2 Direct Air Capture and Utilization in Low-Purity CO_2 Processes.* 2023 AIChE Annual Meeting, Orlando, FL, November 7, 2023 (Oral Presentation)

[4] **Vitor Gama**, San Dinh, Oishi Sanyal, Fernando V. Lima. *Feasibility Analysis of a Membrane System for Direct Air Capture of CO_2.* 2022 AIChE Annual Meeting, Phoenix, AZ, November 17, 2022 (Oral Presentation)

[5] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Designing a Membrane-based Platform for CO_2 Direct Air Capture Systems Analysis.* Gordon Research Conference on Chemical Separations, Ventura, CA, October 2–7, 2022 (Poster Presentation)

Research Experience

West Virginia University Morgantown, WV, USA
Graduate Research Assistant (Ph.D.)[Advisors – Dr. Fernando V. Lima/Dr. Oishi Sanyal] *Aug. 2021 – Currently*

- Simulation and Operability Analysis of gas separation membranes for Direct Air Capture (DAC) with Carbon Capture Utilization and Storage (CCUS).

Federal University of Campina Grande Campina Grande, Paraiba, Brazil
Graduate Research Assistant (M.Sc.) [Advisor - Dr. Antonio Tavernard] *Sept. 2018 – April 2021*

- M.Sc. thesis: "CostApp: a Cost Estimation Tool Developed Using C# and WPF for the Chemical Engineering Field": Developed a Windows OS application for chemical industry equipment cost estimation based on literature models.

Federal University of Campina Grande Campina Grande, Paraiba, Brazil
Developer *Sept. 2019 – April 2021*

- Technological Project for the Development of a Method for Synthesis of Control structures (PETROBRAS/CENPES/PAQTCPB/LENP/UFCG): An automated software capable of easily selecting the most promising self-optimizing control structures in industrial processes.
- Worked on coding the PID routines to control simulated processes uploaded to the BRPWC tool
- Worked on developing mock-ups for the process control user interface of the tool and code troubleshooting

Skills

Programming: Python, MATLAB, Markdown, C#, Javascript, LaTeX and exposure to Java

Technologies/Platforms: Git, GitHub, GitLab

Process simulation: Aspen Plus, Aspen Custom Modeler, HYSYS, AVEVA Process Simulation, PRO/II, ChemCad

Languages: English and Portuguese

Relevant Coursework

Major coursework: Transport Phenomena, Advanced Chemical Engineering Thermodynamics, Chemical Reaction Engineering, Mathematical Methods in Chemical Engineering, Teaching Practicum

Minor coursework: Dynamic Simulations, Membrane Separations, Artifical Intelligence Techniques

Professional leadership and Service

Brazilian Student Association (President) *September 2022 – September 2024*

Statler College DEI Student Committee *May 2022 - May 2023*

CODES Research Group Leader *Sept. 2022 - Sept 2023*

CliftonStrengths

Arranger | Positivity | Input | Woo | Communication